

Solution Requirements

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Functional Requirements

ID	Requirement	Priority
FR-1	System shall load and read the applicant dataset	High
FR-2	System shall preprocess data (handle missing values, encode, scale)	High
FR-3	System shall train multiple ML models for comparison	High
FR-4	System shall evaluate models using standard classification metrics	High
FR-5	System shall accept user input via a web form	High
FR-6	System shall return a loan eligibility prediction in real time	High
FR-7	System shall display the result in a clear, readable format	Medium

Non-Functional Requirements

ID	Requirement	Priority
NFR-1	Predictions shall be returned within 2-3 seconds	High
NFR-2	Interface shall be simple enough for non-technical users	High
NFR-3	Model results shall be consistent and reproducible	Medium
NFR-4	Codebase shall be modular to allow swapping models easily	Medium

Assumptions

- Dataset used is representative of real-world loan applicants.
- This is an academic prototype, not a production banking system.

Constraints

- Limited project timeline (single semester / hackathon-scale effort).
- Small team size; development split between 2 active contributors.